UNIVERSAL GRAVITATION AND SATELLITES IN CIRCULAR ORBIT

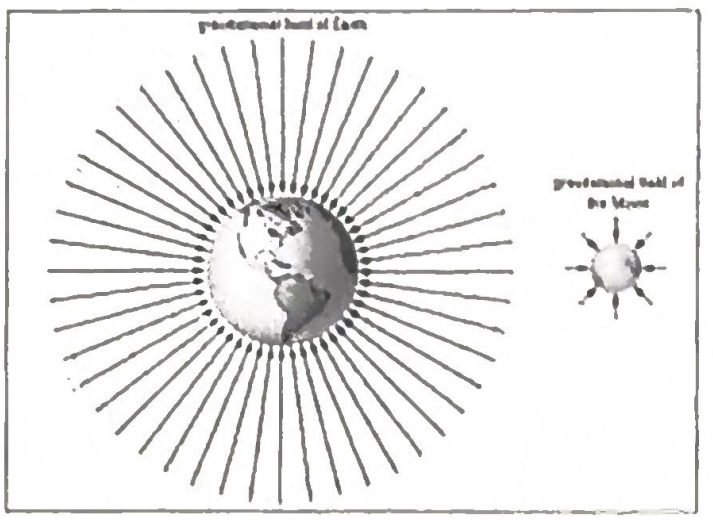
LEARNING GOALS

Students will:

- State Newton's Law of Universal Gravity.
- Solve problems with Newton's Law of Universal Gravity.
- Be able to analyze satellite motion as an application of centripetal acceleration.

NEWTON'S LAW OF UNIVERSAL GRAVITATION

Describing gravity accurately was one of Newton's greatest achievements. Gravitational force, $\vec{F}_G$, is the force of attraction between two objects. All objects in the universe exert gravitational force on all other objects in the universe. Larger objects exert much more force than smaller objects. Gravity is an action-at-a-distance force; stronger when distances between objects are smaller. Fields are used to describe action-at-a-distance forces such as gravity, magnetism, and electrostatics.



A two-dimensional representation of Earth's and the Moon's gravitational fields

Since every object exerts a gravitational force in all directions, every object also has a gravitational field. A test mass can be used to determine the field strength of an planet, $\vec{g}$.

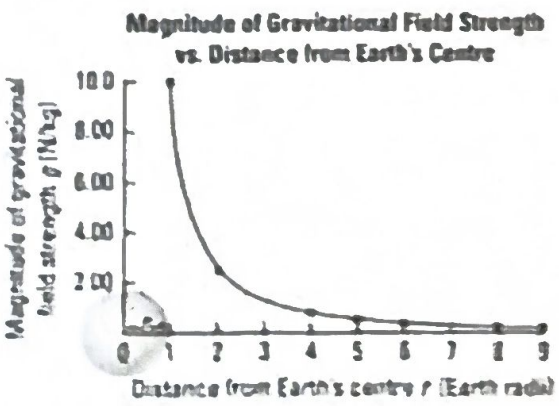
$$\vec{g} = \frac{\vec{F}_g}{m_{test}}$$

Rearranging this equation gives $F_g = mg$, the equation you are familiar with for calculating force of gravity on the surface of the earth.

However, the magnitude of the Earth's gravitational field strength decreases as you move farther from Earth's surface, so g is not always 9.81 N/kg. A more complete understanding of gravitational forces is needed.

Newton hypothesized that the gravitational force exerted by one object on another is directly proportional to the product of their masses.

$F_g \propto m_A m_B$



He also hypothesized that the gravitational force and distance show an inverse square relationship.

$$F_g \propto \frac{1}{r^2}$$

From these two relationships, Newton created his law of universal gravitation.

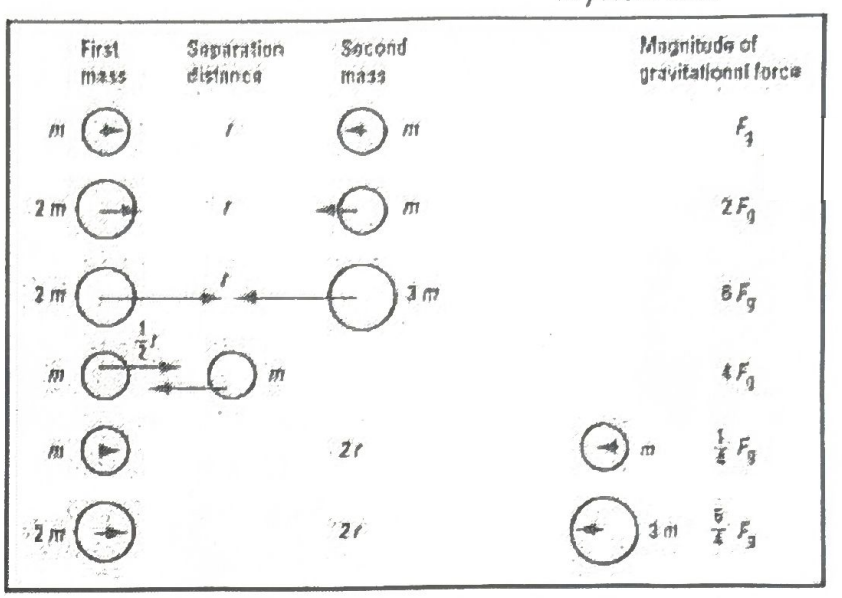
$$F_g \propto \frac{m_A m_B}{r^2}$$

The magnitude of the gravitational force between any two objects can be written as

$$F_g = \frac{G m_A m_B}{r^2}$$

where G is the universal gravitational constant.

CONCEPTS OF GRAVITY



1. As a star ages, it is believed to undergo a variety of changes. One of the last phases of a star's life is to gravitationally collapse into a black hole. What will happen to the orbit of the planets of the solar system if our star (the Sun shrinks into a black hole)? (And of course, this assumes that the planets are unaffected by prior stages of the Sun's evolving stages.)

The orbits stay the same if the Sun's mass stays the same. The distance from its centre is unchanged.

2. Having recently completed his first Physics course, Noah Formula has devised a new business plan based on his teacher's *Physics for Better Living* theme. Noah learned that objects weigh different amounts at different distances from Earth's center. His plan involves buying gold by the weight at one altitude and then selling it at another altitude at the same price per weight. Should Noah buy at a high altitude and sell at a low altitude or vice versa?

Noah should buy at a high altitude because gravity is weaker there, so the gold weighs less. The same gold weighs more at a low altitude, so he should sell it there.

3. Anita Diet is very concerned about her weight but seldom does anything about it. After learning about Newton's law of universal gravitation in Physics class, she becomes all concerned about the possible affect of a change in Earth's mass upon her weight. During a (rare) free moment at the lunch table, she speaks up "How would my weight change if the mass of the Earth increased by 10%?" How would you answer Anita?

Her weight increases by 10% if Earth's radius stays the same, since the force of gravity is proportional to Earth's mass.

4. When comparing mass and size data for the planets Earth and Jupiter, it is observed that Jupiter is about 300 times more massive than Earth. One might quickly conclude that an object on the *surface* of Jupiter would weigh 300 times more than on the surface of the Earth. For instance, one might expect a person who weighs 500 N on Earth would weigh 150000 N on the *surface* of Jupiter. Yet this is not the case. In fact, a 500-N person on Earth weighs about 150 N on the *surface* of Jupiter. Explain how this can be.

Jupiter's larger radius puts you farther from its centre.
Weight depends on both mass and radius: $F_g = GMm/r^2$.
$F_{g,J}/F_{g,E} \approx 300/11^2 \approx 2.5$.
500 N becomes about 1250 N. The printed 150 N is incorrect.

DETERMINING THE VALUE OF *G*

Newton did not discover how to measure *G*. Henry Cavendish, 70 years after Newton's death, used a torsion balance to measure and determine the value of *G*. A torsion balance is used to measure very minute forces. He found that $\boldsymbol{G = 6.67 \times 10^{-11}\ Nm^2/kg^2}$.

FORCE OF GRAVITY ON THE SURFACE OF THE EARTH

For most of the problems where you have had to calculate the force of gravity, you have found the force between the earth and an object on (or very close to) the surface of the earth. In this case, fill in the values using Table 6 on page 710 of your text:

m_{earth} = 5.9722×10^{24} kg r = 6 371 000 m

Use these values to find a short-cut to calculating the force of gravity on an object of mass m_o when it is on the surface of the earth.

$$F_g = \left(\frac{G M_E}{r_E^2}\right) m_o = g m_o$$

$$g = \frac{(6.67 \times 10^{-11})(5.9722 \times 10^{24})}{(6.371 \times 10^6)^2} \approx 9.81 \text{ N/kg}$$

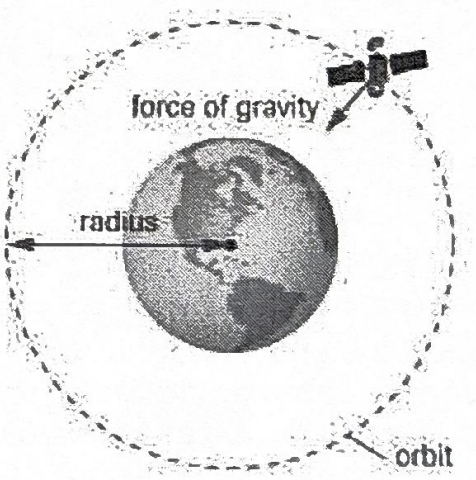
This is the formula you are familiar with from the other problems you have solved. You know it as $F_g = mg$ *where* $g = 9.8$ *N/kg.* If you move away from the earth so that r is no longer the radius of the earth, or if you go to another planet so that r and m_{planet} both change, then the constant value is no longer 9.8 N/kg. However, we can still use the formula $F_g = mg$ if we define the gravitational field strength to be

$$g = \frac{Gm_{planet}}{r^2}$$

and calculate a value of g for wherever we are.

SATELLITES AND SPACE STATIONS

A **satellite** is an object or a body that revolves around another object, which is usually much larger in mass. A space station is a type of artificial satellite in which people live and work.



a) For any satellite, what is the force that keeps it in its orbit?

Force of Gravity ($\vec{F_g}$)

b) To analyze the motion of a satellite, we combine Newton's Law of Universal gravitation with the equations for centripetal force. Sub in the **equations** for the two forces below:

$$F_c = F_G$$

$$\frac{mv^2}{r} = \frac{G M_E m}{r^2}$$

c) Starting with your equation above, derive an expression for the speed of the satellite orbiting the earth.

$$v^2 = \frac{G M_E}{r}$$

$$v = \sqrt{\frac{G M_E}{r}}$$

d) Derive an expression for the radius of a satellite's orbit around Earth in terms of the **period** of revolution using the appropriate expression for the centripetal acceleration.

$$\frac{GM_E}{r^2} = \frac{4\pi^2 r}{T^2}$$

$$r^3 = \frac{GM_E T^2}{4\pi^2}$$

$$r = \sqrt[3]{\frac{GM_E T^2}{4\pi^2}}$$

EXAMPLE: HUBBLE TELESCOPE

The Hubble Space Telescope is in a roughly circular orbit 598 km above the Earth's surface. Find (a) its speed in orbit; (b) its orbital period. HINT: you must consider the centre-to-centre distance between the Earth and the telescope.

$$r = R_E + h = 6.969 \times 10^6 \text{ m}$$

$$v = \sqrt{\frac{GM_E}{r}}$$

$$v = \sqrt{\frac{(6.67 \times 10^{-11})(5.9722 \times 10^{24})}{6.969 \times 10^6}}$$

$$v \approx 7.56 \times 10^3 \text{ m/s}$$

$$T = \frac{2\pi r}{v}$$

$$T = \frac{2\pi (6.969 \times 10^6)}{7560.4}$$

$$T \approx 5.79 \times 10^3 \text{ s}$$

CONTINUOUS FREEFALL

When the space station (or any satellite) orbits the earth, it travels at the appropriate speed that keeps it approximately the same distance from Earth's centre. As it circles Earth, it is undergoing **constant free fall** toward Earth.

a) A satellite is "always falling, but missing the earth". Explain what this means in terms of orbits.

Gravity pulls the satellite inward. It moves sideways fast enough that Earth curves away beneath it as it falls.

b) How do you think an astronaut on the ISS would feel based on this idea?

The astronaut feels weightless because the astronaut and the ISS fall together, so $F_N = 0$.

ANSWERS

c. $v = \sqrt{\frac{Gm_E}{r}}$ 2. d= $\sqrt[3]{\frac{Gm_E T^2}{4\pi^2}}$ Example: a) 7559 m/s b) 5793 s

CLASS WORK

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